

Carl Paolino

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science, Minor in Mathematics

August 2023 – May 2027

- Concentration in Intelligence and Information Internetworks
- Coursework: Data Structures & Algorithms, Computer Organization, Design & Analysis of Algorithms, SQL, Artificial Intelligence, Computer Networking, Linear Algebra, Object Oriented Programming

EXPERIENCE

AI Engineer Intern

January 2026 – August 2026

Georgia Tech Research Institute (GTRI)

Atlanta, GA

- Designing and implementing an end-to-end machine learning pipeline to automate unit test generation using Ollama-hosted LLMs, remotely optimizing models on GPU-enabled infrastructure for scalable inference
- Constantly Containerizing the pipeline using Docker and integrated LangChain and LlamaIndex to orchestrate prompt workflows, manage code reviews, and apply automated test generation across multiple programs

Software Engineer Contractor

October 2025 – December 2025

Foundry AI

Atlanta, GA

- **Government Contract**
 - * Implemented an AI testing pipeline for unit, integration, and end-to-end tests with CI integration, automated LLM-as-Judge scoring, and budget-aware execution to achieve $\geq 90\%$ accuracy on test benchmark
 - * Developed an AI-powered conversational pipeline with a live dashboard and chatbot interface, allowing users to query data and receive real-time, context-aware responses generated through SQL and LLM inference
- **Commercial IOS Application**
 - * Developed an AI-powered native IOS application, implementing a robust tool orchestration framework to manage model-driven actions, data flows, and device capabilities
 - * Designed and enforced a permission management system, ensuring secure execution of AI tools, user data isolation, and compliance with platform security best practices

Software Developer Intern

July 2025 – August 2025

Healthmap Solutions

Tampa, FL

- Developed an internal web tool with Python, React, and Django to serve as a test harness for health analytics and care coordination
- Learned about deployment to a serverless application using AWS Lambda connected to an RDS database, with automated CI/CD pipelines through GitHub Actions

Undergraduate Researcher

January 2025 – May 2026

Georgia Institute of Technology

Atlanta, GA

- Improving the design and implementation of a forest-fire prevention drone communication system
- Developing a Google Cloud-based data pipeline connecting a Raspberry Pi drone system to Firestore, enabling real-time telemetry and live video streaming via Cloud Run and Pub/Sub

PROJECTS

Trade Mind | *Next.js, FastAPI, Ollama, vLLM, MLflow, Prometheus, Pinecone*

March 2026 – Present

- Building an AI investment intelligence platform trained on financial news to generate portfolio recommendations
- Fine-tuning a 7B open-source LLM behind a RAG pipeline using embeddings and hybrid search
- Containerizing microservices on Kubernetes with autoscaling, JWT auth and rate-limiting on inference endpoints

CS 3600 | *NumPy, Pandas, Python, TensorFlow, PyTorch*

May 2025 – July 2025

- Developed goal-state detection, successor functions, and heuristic evaluation for agents to explore and plan
- Explored reinforcement learning algorithms to solve Markov Decision Processes
- Implemented Q-Learning algorithm to train autonomous agents in TextWorld-Express environments

TECHNICAL SKILLS

Languages: Java, Python, C, JavaScript, SQL, TypeScript, Swift

DevOps and & Analytics: CI/CD, Remote SSH, Git, GitLab, Docker, Pandas, NumPy, Excel

Developer Tools: Git/GitHub, Linux, Bash, SDLC, Scrum, Google Cloud Console, AWS, PostgreSQL, MongoDB

Frameworks/Libraries: React, Ollama, LLama-index, LangChain, TensorFlow, Scikit-learn, SwiftUI, Django